

KHUVSGUL, Mongolia

WMO: 443860

Lat: 43.6044N

Lon: 109.6361E

Elev: 3264

StdP: 13.04

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-11.3	-23.5	1.7	-3.9	-19.1	2.2	-4.3	31.9	18.3	29.6	20.7	3.5	320	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	94.9	67.0	92.0	65.8	89.0	64.8	73.0	88.8	70.6	85.9	68.3	83.2	10.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	116.4	83.6	65.3	105.6	80.0	63.0	97.4	78.0	39.3	89.3	36.8	86.2	34.8	83.6	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.9	27.6	23.7	-19.6	100.2	5.6	3.0	-23.6	102.3	-26.9	104.1	-30.1	105.8	-34.2	108.0		
			-20.1	73.9	5.7	5.2	-24.2	77.6	-27.6	80.6	-30.8	83.5	-34.9	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	6.2	14.0	31.8	47.9	60.4	71.6	77.5	74.0	62.0	45.7	25.3	9.4	
	DBStd	26.44	9.31	10.78	10.94	9.86	9.07	6.84	5.36	6.13	8.11	9.31	10.45	9.12	
	HDD50	5311	1358	1009	571	153	17	0	0	0	11	191	742	1259	
	HDD65	8686	1823	1429	1030	517	199	19	1	7	146	599	1192	1724	
	CDD50	3110	0	0	5	90	339	649	853	744	371	59	0	0	
	CDD65	1011	0	0	0	3	56	219	389	285	57	2	0	0	
	CDH74	11003	0	0	0	67	787	2385	4355	2867	523	19	0	0	
	CDH80	4522	0	0	0	10	269	969	1989	1166	117	2	0	0	
Wind	WSAvg	9.6	7.9	8.8	10.4	11.3	12.8	10.1	9.1	8.4	8.6	9.6	9.1	8.6	
Precipitation	PrecAvg	5.1	0.1	0.1	0.2	0.1	0.4	0.8	1.3	1.2	0.5	0.3	0.2	0.1	
	PrecMax	8.5	0.2	0.5	0.6	0.4	1.7	1.4	3.4	3.1	1.4	0.9	0.5	0.3	
	PrecMin	3.1	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.0	0.0	0.0	0.0	0.0	
	PrecStd	1.3	0.0	0.1	0.2	0.1	0.4	0.4	0.6	0.8	0.4	0.2	0.1	0.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	47.3	66.6	81.8	92.0	97.1	100.4	97.4	87.1	76.7	56.3	38.4	
		MCWB	26.2	36.6	44.7	54.3	57.5	67.3	72.2	69.3	61.1	55.9	42.7	31.5	
	2%	DB	29.3	41.2	59.9	75.3	86.8	92.8	95.8	93.3	83.5	70.4	50.3	30.8	
		MCWB	23.1	32.5	41.7	50.9	57.0	64.4	68.4	67.7	60.9	52.4	39.6	25.3	
	5%	DB	24.3	36.0	54.6	70.1	82.4	89.4	92.7	89.9	80.2	65.4	44.8	26.4	
		MCWB	19.6	28.4	38.8	48.5	55.4	63.6	67.2	65.9	59.1	49.5	35.7	21.8	
	10%	DB	20.1	30.9	49.7	65.2	78.0	85.7	89.6	86.5	76.6	60.6	40.3	22.7	
		MCWB	16.4	24.5	35.9	46.1	53.6	61.5	66.6	64.5	57.1	46.5	32.6	19.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	38.5	47.0	58.0	63.6	73.0	79.8	74.5	68.4	58.9	43.9	30.9
			MCDB	34.4	46.0	60.9	78.1	84.0	94.2	97.6	91.5	82.4	72.8	54.5	38.4
2%		WB	23.2	32.6	42.5	52.7	59.6	68.8	74.7	71.7	64.5	54.3	40.1	25.7	
		MCDB	29.1	40.6	59.2	70.4	79.1	86.3	90.2	86.5	78.7	66.9	49.8	30.7	
5%		WB	19.8	28.8	39.3	49.8	57.3	66.6	71.8	69.6	61.7	50.2	36.4	22.0	
		MCDB	24.3	36.1	53.4	67.2	78.2	82.7	86.8	85.5	73.5	63.5	44.6	26.0	
10%		WB	16.5	25.0	36.4	47.1	55.1	64.5	69.6	67.2	59.1	47.2	33.2	19.0	
		MCDB	20.1	31.2	49.0	64.7	74.7	80.3	84.0	82.0	72.9	59.1	40.3	22.6	
Mean Daily Temperature Range	5% DB	MDBR	19.0	21.7	23.4	22.6	23.4	20.8	19.7	20.0	20.8	20.6	18.7	17.2	
		MCDBR	22.2	26.4	28.0	28.4	28.2	24.6	23.2	23.2	24.3	24.5	22.7	20.3	
		MCWBR	17.6	19.4	17.1	15.8	13.8	11.2	9.2	10.4	11.9	14.2	15.1	15.8	
	5% WB	MCDBR	22.0	25.9	27.3	26.1	26.4	21.1	20.0	21.0	21.1	23.6	22.1	19.8	
		MCWBR	17.7	19.4	17.3	16.1	15.3	11.6	11.2	12.0	13.2	15.3	15.5	15.8	
Clear-Sky Solar Irradiance	taub	0.274	0.304	0.346	0.398	0.438	0.462	0.437	0.416	0.382	0.334	0.292	0.273		
	taud	2.265	2.190	2.104	2.011	1.953	1.976	2.113	2.148	2.168	2.236	2.263	2.244		
	Ebn at Noon	270	279	280	273	265	259	264	266	265	264	259	257		
	Edh at Noon	27	35	44	52	57	56	49	45	41	33	27	25		
All-Sky Solar Radiation	RadAvg	703	1055	1557	1956	2200	2176	2094	1924	1611	1212	810	595		
	RadStd	82	100	86	90	98	106	90	75	64	47	50	63		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	-2.05	N/A	-1.03	-1.35	N/A	N/A	N/A	N/A	

Nomenclature: See separate page